

DATA420-25S1 (C)
Assignment 1
GHCN Data Analysis using Spark
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1. Background

1.1 Purpose of the Study

This assignment used Apache Spark to analyse climate data from the Global Historical Climatology Network (GHCN). The dataset spans over 250 years and have records over 100,000+ stations across 200 countries. Spark helps to process this large-scale data and helps to create visualizations to understand climate trends.

1.2 Overview

The GHCN-Daily dataset contains daily weather data like maximum and minimum temperature (TMAX, TMIN), rainfall (PRCP), snowfall (SNOW), and snow depth (SNWD). This data is saved in CSV format. Extra information about the stations, countries, and recorded elements is stored in separate files with fixed column widths. All the data is stored in Azure Blob Storage, where we can read input files from a shared folder and save our results in a personal folder. Using Apache Spark, we clean, join, and analyse this large dataset to answer questions about the weather and create charts to help understand long-term climate patterns around the world.

1.3 Challenges Faced

The dataset size over 3 billion records made processing slow, it required efficient Spark operations. Metadata files were in fixed-width format, makes column extraction tricky. Large tables joining needed careful memory management. Careful section of spark Operations is quiet challenging, stations had missing data, which had to be handled during analysis. While plotting choropleth, need of conversion from country name to ISO3.

After completion of this assignment, I have learned how to deal big data, optimize Spark queries, and how visualize climate trends effectively using pandas.

2. Processing

In this section, I prepared the GHCN-Daily and metadata files for analysis. By using HDFS commands to check folder structures and file sizes in Azure Blob Storage. Then I used schemas to load data into Spark and used `substring()` to read the fixed-width format of meta data files. Then, I joined the station data with country, state, and inventory info makes enriched station data.

2.1 Dataset Structure

2.1.1 Files and Descriptions

The GHCN data is stored in Azure Blob Storage at:
`wasbs://campus-data@madstorage002.blob.core.windows.net/ghcnd/`

The `ghcnd/` folder contains five items:

File Name	Size	Description
daily/ (264 files)	~13.02 GB	Compressed CSVs (one per year from 1750–2025, excluding 1751–1762). Contains daily weather observations.
ghcnd-stations.txt	~10.63 MB	Metadata for all weather stations.
ghcnd-inventory.txt	~33.59 MB	Records availability of each element per station.
ghcnd-countries.txt	~3.6 KB	Country codes and names.
ghcnd-states.txt	~1.1 KB	U.S. state codes and names.

Table 2.1.1: Files in the GHCN Dataset Folder

2.1.2 Directory Tree View

wasbs://campus-data@madssstorage002.blob.core.windows.net/ghcnd/

```
|-----daily/ (13.02 GB, 264 files)
|       |----1750.csv.gz (1.38 MB)
|       |----1763.csv.gz (3.36 KB)
|       |----1764.csv.gz (3.33 KB)
|       |----1765.csv.gz (3.34 KB)
|       |---- ...
|       |----2024.csv.gz (168.5 MB)
|       |----2025.csv.gz (17.06 MB)
|
|-----ghcnd-inventory.txt (33.59 MB)
|-----ghcnd-stations.txt (11.15 MB)
|-----ghcnd-countries.txt (3.66 KB)
|-----ghcnd-states.txt (1.1 KB)
```

2.1.3 Yearly File Size Trends

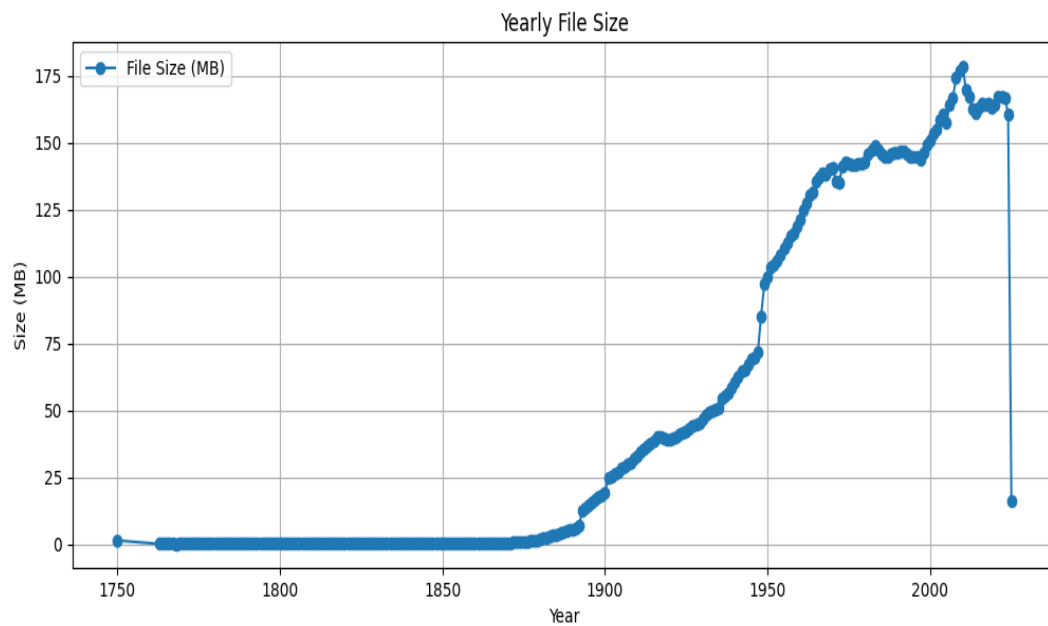


Figure 2.1.3: Yearly File Size Trends (1750–2025)

Year Range	Approx. Size Trend	Interpretation
1750	~1.38 MB	Might be some consolidated records
1751-1762	No data	Missing values
1763–1800	~3.28 KB to ~8 KB	Very minimum data
1800–1850	~8 KB to ~50 KB	Gradual increase in records. Still sparse global coverage.
1850–1900	~50 KB to ~20 MB	Significant growth, especially post-1860s. Better recordings.
1900–1950	~20 MB to ~100 MB	Steady increase
1950–2000	~100 MB to ~150 MB	Recording grow rapidly.
2000–2023	~150 MB to ~165 MB	Stable, high-volume data.
2025	~16 MB	Incomplete data for this year (still ongoing).

Table 2.1.3: Interpretation of Yearly File Size Graph

This graphs clearly shows increase in file size over years, which means expansion of global climate data collection.

2.1.4 Summary Table: Dataset Size

The total file size is approximately 13.06 GB, from this the daily alone is 13.02 GB, which is almost 99% of the total size. Meta data files are very small to compare daily:

Dataset Name	File Size (Bytes)	File Size (MB)	File Size (GB)
ghcnd-inventory.txt	35,218,290	33.59	0.0328
ghcnd-stations.txt	11,150,502	10.63	0.0104
ghcnd-countries.txt	3,659	0.0035	0.000003
ghcnd-states.txt	1,086	0.0010	0.000001

daily	13,975,887,693	13,328.45	13.02
-------	----------------	-----------	-------

Table 2.1.4: Summary Table

2.2 Loading and Schemas

2.2.1 Schema Definition for daily Dataset

A schema was defined using `pyspark.sql.types.StructType` using the following fields and data types:

Column Name	Data Type	Reason
ID	StringType()	Station codes are alphanumeric.
DATE	DateType()	Needed for filtering and time-series operations (converted from string).
ELEMENT	StringType()	Weather variable (e.g., TMAX, PRCP, etc.).
VALUE	FloatType()	For performing numerical analysis.
MEASUREMENT_FLAG	StringType()	Categorical field with missing values allowed.
QUALITY_FLAG	StringType()	Used to filter poor-quality data.
SOURCE_FLAG	StringType()	Identifies source of the data.
OBSERVATION_TIME	StringType()	The time of observation keep as string type not used anywhere.

Table 2.2.1: Schema Definition

2.2.2 Loading the daily Data (Most Recent Year)

The 2024 file was loaded with the above schema. Initial issues:

- Spark read all columns as `StringType` when schema wasn't applied.
- `DATE` needed conversion to `DateType` using `to_date()`.

- VALUE was cast from string to FloatType.
- Flags missing values need to handle.

2.2.3 Loading Fixed-Width Metadata Files

Stations, states, countries, and inventory datasets are of fixed-width format, each of this read by `spark.read.text()` and parsed using `pyspark.sql.functions.substring()` for extract columns.

Example (for stations):

```
substring("value", 1, 11).alias("ID")
```

- Extract from 1st to 11th character
- This method was used for all metadata datasets by specifying their column ranges.

2.2.4 Data Type Summary

Dataset	Loading Method	Notes
daily	<code>read.csv()</code> with schema	DATE and VALUE required type casting.
stations	<code>read.text()</code> + <code>substring()</code>	ID, LAT/LONG, STATE, NAME, etc.
states	<code>read.text()</code> + <code>substring()</code>	Two fields: CODE and NAME.
countries	<code>read.text()</code> + <code>substring()</code>	Two fields: CODE and NAME.
inventory	<code>read.text()</code> + <code>substring()</code>	ID, ELEMENT, FIRSTYEAR, LASTYEAR, LAT/LONG.

Table 2.2.4: Data Type Summary

2.2.5 Extra Processing Required

- Used `substring()` to parse fixed-width formats.
- Cast DATE to `DateType` and VALUE to `FloatType`.
- Replaced blank strings with null using
`when(trim(col(c)) == "", None).otherwise(col(c))`
- Allowed nullable fields for handling missing values.

2.2.6 Row Counts of Metadata Tables

Dataset	Number of Rows
inventory	765,615
stations	129,657
countries	219
states	74

Table 2.2.6: Row Counts of Meta Data Tables

2.3 Enrichment of Station Metadata

To enrich the station metadata, performed a series joins across: stations, countries, states, and inventory.

2.3.1. Joins and Metadata Enhancement

Operation	Technical Implementation & Description
Extracted country code	- Used <code>pyspark.sql.functions.substring()</code> to extract the first two characters from the ID column: <i>stations = stations.withColumn("COUNTRY_CODE", substring("ID", 1, 2))</i> - This new column was used to link with the countries dataset.
Join with countries	- Performed a LEFT JOIN on COUNTRY_CODE using: <i>stations = stations.join(countries, on="COUNTRY_CODE", how="left")</i> - This brought in the COUNTRY_NAME column from the countries dataset into stations.
Join with states (US only)	- Performed a LEFT JOIN: - Join only if STATE is not null (i.e., US stations) <i>stations = stations.join(states, on="STATE_CODE", how="left")</i>

	Non-US rows remain as NULL values in STATE_NAME.
Inventory summary	- Computed each station's active period and total element count: <code>inventory_summary = inventory.groupBy("ID").agg(min("FIRSTYEAR").alias("FIRSTYEAR"), max("LASTYEAR").alias("LASTYEAR"), countDistinct("ELEMENT").alias("TOTAL_ELEMENTS"))</code> - The earliest and latest year of recorded. - Count how many unique elements.
Core and other elements count	- Created a binary flag for core elements (PRCP, TMAX, TMIN, SNOW, SNWD) using: <code>inventory = inventory.withColumn("CORE_ELEMENT", when(col("ELEMENT").isin(core_list), 1).otherwise(0))</code> - Then grouped and aggregated: <code>station_element_types = inventory.groupBy("ID").agg(sum("CORE_ELEMENT").alias("CORE_ELEMENTS"), (countDistinct("ELEMENT") - sum("CORE_ELEMENT")).alias("OTHER_ELEMENTS"))</code> - Count how many core elements.
Final enrichment (metadata + inventory)	- First joined inventory_summary with station_element_types: <code>inventory_enriched = inventory_summary.join(station_element_types, on="ID", how="left")</code> - Then merged with the stations to create the final enriched dataset: <code>enriched_stations = stations.join(inventory_enriched, on="ID", how="left")</code> - This gets a fully enriched station metadata table with: - Location info (country, state) - Station name - Time range of operation - Core and other element counts

Table 2.3.1: Process of Joining tables

2.3.2 Key Metrics from Enriched Data

Metric	Value
Stations collecting all 5 core elements	20,504
Stations collecting only PRCP	16,313

Table 2.3.2: Key Metrics from Enriched Data

2.3.3 Final Enriched Schema Overview

Column Name	Description
ID	Unique station identifier
STATE_CODE	US state code (nullable)
COUNTRY_CODE	Extracted from station ID
LATITUDE	Geographic coordinate
LONGITUDE	Geographic coordinate
ELEVATION	Station elevation in meters
NAME	Station name
GSN_FLAG	Global Summary of the Day network flag
HCN_CRN_FLAG	Historical Climatology/Climate Reference Network flag
WMO_ID	World Meteorological Organization ID
COUNTRY_NAME	Joined from countries dataset
STATE_NAME	Joined from states dataset
FIRSTYEAR	First year the station collected any element
LASTYEAR	Last year of collection
TOTAL_ELEMENTS	Total unique weather elements collected
CORE_ELEMENTS	Count of core elements collected
OTHER_ELEMENTS	Count of non-core elements

Table 2.3.3: Final Enriched Schema Overview

2.3.4 Output Format

The final enriched_stations DataFrame was saved to Azure Blob Storage using the Parquet format:

Reason for choosing Parquet:

- Columnar storage -> faster queries in Spark
- Compressed format -> reduced storage cost
- Preserves schema and data types efficiently

Saved at

wasbs://campus-

user@madsstorage002.blob.core.windows.net/asa349/enriched_stations.parquet

2.4 Verifying Missing Stations in Daily Dataset

The goal was to identify whether any station IDs in the daily weather dataset were not present in the enriched stations metadata table built in above.

2.4.1 Methodology

Two different join operations were used in Spark to solve this task (Performance Comparison):

Approach 1: LEFT OUTER JOIN

- I performed a LEFT JOIN between the 2025 daily sample and the enriched station metadata on the ID column:

```
daily_enriched = daily_sample.join(enriched_stations, on="ID", how="left")
```

- I filtered rows where no match was found in the metadata (i.e., COUNTRY_CODE was null):

```
missing_stations =
```

```
daily_enriched.filter(col("COUNTRY_CODE").isNull()).select("ID").distinct()
```

Approach 2: LEFT ANTI JOIN (Optimized Method)

To improve efficiency, I filtered only distinct station IDs from both datasets and applied a LEFT ANTI JOIN:

```
missing_ids = daily_ids.join(station_ids, on="ID", how="left_anti")
```

2.4.2 Results and Comparison

Both methods yielded the same result:

Number of stations in daily that are not in stations: 0

Performance Comparison

Feature	LEFT JOIN	LEFT ANTI JOIN
Join Type	LeftOuter	LeftAnti
Data Scanned	All columns from both datasets	Only ID columns
Memory Usage	High	Low
Filtering	Performed after join (using isNull)	Done during join itself
Speed	Slower due to more data movement	Faster and more efficient
Physical Plan	Broadcast + HashJoin + Filter	Broadcast + HashJoin on ID only

Table 2.4.2: Performance Comparison

- After this finding, for full daily dataset (~13 GB), I used a LEFT ANTI JOIN to check if any station IDs were missing from the enriched_stations table.
- Result: All station IDs matched 0 missing stations.
- Findings: For large dataset a LEFT JOIN would be slower and use more memory. The LEFT ANTI JOIN is faster and more efficient.

3. Analysis

After preparing and enriching datasets, this section mainly focuses on some key questions answering about weather stations and daily observations using PySpark. I was used Spark DataFrame operations such as filter(), groupBy(), agg(), join(), collect_set() and user-defined functions for this analysis.

3.1 Station Metadata Analysis

3.1.1 Total Stations

There are **129,657 unique weather stations** recorded in the dataset.

3.1.2 Active Stations in 2025

For finding which are the stations active in 2025, I first used the enriched station metadata and filter LASTYEAR == 2025. Then, cross-check the results by looked at the daily dataset for 2025.

Step	Description	Stations Count
Metadata Filtering	Filtered enriched_stations where LASTYEAR == 2025	30,735
Daily Dataset Extraction	Extracted distinct station IDs from 2025 daily dataset using: select("ID").distinct().count()	32,224
Identify Mismatches	Performed left anti join to find station IDs in daily but not in enriched stations data.	1,489 (unmatched stations)
Investigate Missing Stations	Joined unmatched IDs with enriched stations to find missing countries	1,475 from US; few from Canada, Ethiopia, Fiji, etc.

Table 3.1.2.1: Steps to find Active Stations in 2025

Extra Stations in Daily Dataset:

Country	Count
United States	1,475
Canada	4

Ethiopia	3
Puerto Rico (US)	2
Russia	2
Paraguay	2
Fiji	1

Table 3.1.2.2: Extra Stations in Daily Dataset:

All stations in full daily dataset were also found in enriched station metadata (Found in Processing Q4), which means meta data covers all stations. But this discrepancy is likely limited to minor timing or update delays in station records for 2025 .

3.1.3 Stations by Network Membership

To determine how many stations belong to each climate monitoring network, I used the GSN_FLAG and HCN_CRN_FLAG columns from the enriched stations dataset. These flags indicate whether a station is part of the:

- GCOS Surface Network (GSN) (Focus on Global)
- US Historical Climatology Network (HCN) (US)
- US Climate Reference Network (CRN) (Modern, US)

Network	Count
GSN	991
HCN	1,218
CRN	234

Table 3.1.3: Stations by Network Membership

15 stations belong to both GSN and HCN networks.

There is no overlap involving CRN and others.

3.1.4 Hemisphere Distribution

To analyse the distribution of weather stations across hemispheres, I filtered the enriched stations dataset based on latitude:

- Stations in the Southern Hemisphere identified where LATITUDE < 0.

- Stations in the Northern Hemisphere has LATITUDE ≥ 0

Hemisphere	Count	Percentage
Southern Hemisphere	25,357	19.6%
Northern Hemisphere	104,300	80.4%

Table 3.1.4: Stations Count over Hemispheres

This shows that, Northern Hemisphere have more weather stations, probably because there is more land and better systems to record weather there.

3.1.5 Stations in US Territories (Excluding Mainland US)

For finding weather stations located in U.S. territories but not in the mainland US, I use the following logic on the COUNTRY_NAME field of enriched_stations:

- Include rows where the Country name **contains** "United States" (e.g., "Puerto Rico [United States]") because territories represent name + [United States].
- But Exclude rows where the name is exactly "United States".
- This selects only territories like Guam, American Samoa, etc.).

PySpark condition used:

```
(trim(col("COUNTRY_NAME")).contains("United States")) &
(trim(col("COUNTRY_NAME")) != "United States")
```

COUNTRY_NAME	Station Count
Puerto Rico [United States]	260
Virgin Islands [United States]	77
Guam [United States]	34
American Samoa [United States]	21
Northern Mariana Islands [United States]	11
Johnston Atoll [United States]	4
Midway Islands [United States]	3
Palmyra Atoll [United States]	3
Wake Island [United States]	1
Total	414

Table 3.1.5: Stations in US Territories

3.1.6 Country and State Station Counts

To better understand the geographic distribution of weather stations, I calculated the total number of stations per country and state using the enriched station metadata.

Stations by Country

Using `groupBy("COUNTRY_NAME", "COUNTRY_CODE")`, I counted the total number of stations in each country. I also counted how many countries got exactly one station.

- Top 3 Countries:
 - United States: 75,846 stations
 - Australia: 17,088 stations
 - Canada: 9,269 stations
- 41 countries have only one weather station, like Niue [New Zealand] and Wake Island [United States], indicate those regions have limited climate monitoring.

The station counts joined to countries table using `COUNTRY_CODE` and saved as a Parquet file for further use.

Stations by State

Same for state, I grouped by `STATE_NAME` and `STATE_CODE` (excluding null values) to count the number of stations in each U.S. state and territory.

- Top 3 States:
 - Texas: 6,472 stations
 - Colorado: 4,784 stations
 - California: 3,166 stations
- Territories with fewest stations included:
 - Northern Mariana Islands and U.S. Minor Outlying Islands (11 each)
 - Palau (13), American Samoa (21), and others

These state-level counts joined to the states metadata and also saved as a Parquet file.

3.2 Geospatial Distance Between Stations

I used a Spark user-defined function (UDF) for Haversine formula to calculate distance between weather stations based on their latitudes and longitudes. This formula considers the

Earth's round shape. I applied it to all pairs of stations in New Zealand using a CROSS JOIN to find the two closest stations.

3.2.1 Calculating Geographic Distance (Haversine Formula)

Formula Concept:

Haversine calculate the shortest distance between two points on a sphere surface using latitude and longitude.

- Earth's radius (R): 6371 km.
- Latitude and longitude: Converted from degrees to radians.

$$a = \sin^2(\Delta\varphi / 2) + \cos(\varphi_1) \times \cos(\varphi_2) \times \sin^2(\Delta\lambda / 2)$$

$$c = 2 \times \text{atan2}(\sqrt{a}, \sqrt{1 - a})$$

$$d = R \times c$$

Where:

$\varphi_1, \varphi_2 = \text{latitudes in radians}$

$\lambda_2, \lambda_1 = \text{longitudes in radians}$

$\Delta\varphi = \varphi_2 - \varphi_1$

$\Delta\lambda = \lambda_2 - \lambda_1$

$R = 6371 \text{ km (average Earth radius)}$

c: Central angle between two points on the surface of the sphere, from earth centre.

d: Actual distance between two points on earth surface

3.2.2 Implementation Steps

1. Define the Haversine function for taking four parameters: latitude and longitude of both stations.
2. Converted degrees latitude and longitude to radians using `math.radians`.
3. Applied the Haversine formula for calculating distances between two coordinates.
4. Using `udf(haversine, DoubleType())` register the function as a Spark UDF allows it used on Spark DataFrames.
5. Filter stations in New Zealand from the enriched metadata.
6. CROSS JOIN on the filtered stations to compare each station with every other.
7. Applied the UDF to compute pairwise distances.
8. Select the closest station pairs by sorting the result by distance and exclude self-pairs.

This method was chosen because it accounts the curved shape of the earth (Wikipedia contributors, n.d.).

3.2.3 Closest Station Pair in New Zealand:

Station 1	Station 2	Distance (km)
Paraparaumu AWS	Wellington Aero AWS	50.53 km

Table 3.2.2.1: Closest Station Pair in New Zealand

Station Name	Latitude	Longitude	Location Context
Wellington Aero AWS	-41.333	174.800	Further south, closer to the coast, more exposed to wind and storms (Cook Strait winds)
Paraparaumu AWS	-40.900	174.983	Slightly north, more inland/sheltered, likely calmer conditions (Kapiti Island shield)

Table 3.2.2.2: Latitude Longitude of Closest Station

Even though the two stations are only about 50 km apart, they might have quite different weather because of their location and surroundings.

3.3 Daily Dataset Analysis

Using PySpark for exploring the full daily dataset, focusing on core elements such as TMAX, TMIN, PRCP, SNOW, and SNWD. I counted how many of each element and how often TMAX appears without TMIN.

I increased Spark resources for faster the processing of daily dataset (~3B rows).

3.3.1 Total Number of Records

- The full daily dataset contains 3,139,143,397 rows.
- This includes all weather observations (across all stations and elements) from 1750 to 2025.
- Due to the dataset's size (~13 GB), I did this without caching or collecting data.

3.3.2 Core Element Counts

I filtered the dataset for including only the five core elements. The total number of records for each element is shown below:

Element	Count
PRCP	1,079,767,077
TMAX	460,114,659
TMIN	458,928,768
SNOW	359,249,644
SNWD	300,711,620

Table 3.3.2: Core Elements Counts

- PRCP (precipitation) has the highest number of observations.

3.3.3 Unmatched Temperature Records

Many stations were reported TMAX and TMIN, but not always together. To identify this issue, I did this:

Step	Operation
Group data	Grouped daily records by ID (station) and DATE.
Collect elements	Use collect_set("ELEMENT") to collect all reported elements (TMAX, TMIN, etc.) for each station-date pair.
Filter TMAX only	Filter rows where elements contains TMAX but not TMIN using array_contains().
Count results	Counted total unmatched TMAX observations using .count().
Count stations	Counted distinct station IDs with unmatched TMAX using .select("ID").distinct().count().

Table 3.3.3: Unmatched Temperature Records -Analysis Steps

- Result: 10,660,214 unmatched TMAX observations.

3.3.4 Unique Stations with Missing TMIN

- A total of 28,754 unique stations have unmatched TMAX records.

Key Conclusions:

- PRCP is the highest recorded element in the dataset.
- A portion of TMAX records have missing corresponding TMIN, which could affect analysis.

Summary Result

Metric	Count
Total Observations (Daily Dataset)	3,139,143,397
PRCP Observations	1,079,767,077
TMAX Observations	460,114,659
TMIN Observations	458,928,768
SNOW Observations	359,249,644
SNWD Observations	300,711,620
TMAX without matching TMIN	10,660,214
Stations with unmatched TMAX	28,754

Table 3.3.4: Summary of Counts

4 Visualizations

This section focuses on key climate trends using time series and geospatial plot. I analyzed TMIN and TMAX for 15 New Zealand stations and visualized national temperature trends. For global level, used yearly rainfall by country and mapped average precipitation in 2024 in choropleth map. For all plotting aggregated data is used for avoiding memory issues.

4.1 TMIN and TMAX Trends at New Zealand Stations

4.1.1 Data Summary for TMIN and TMAX Observations in New Zealand

I filtered the daily data for New Zealand and reshaped it to include both TMIN and TMAX per station and date. The table below shows the counts:

Metric	Count	Description
Original observations	491,441	Total TMIN or TMAX records
Pivoted observations	265,205	Unique station-date rows
TMIN present	237,320	Rows with TMIN
TMAX present	254,121	Rows with TMAX
TMIN missing	27,885	Rows missing TMIN
TMAX missing	11,084	Rows missing TMAX
Both TMIN and TMAX present	226,236	Complete rows with both values

Table 4.1.1: TMIN and TMAX Count

4.1.2 Years of Data Coverage

For New Zealand TMIN and TMAX data spans from 1940 to 2025, covered a total of 86 years. There are no missing years in this period.

Description	Value
Years covered	1940 to 2025
Total years	86
Missing years	None

Table 4.1.2: Years of Data Coverage

4.1.3 Number of Unique Stations

The dataset contains 15 unique weather stations from New Zealand, from both main cities (like Auckland, Wellington, and Christchurch) and remote sites (like Campbell Island and Raoul Island).

Sl. No.	Station Name	Station ID
1	KAITAIA	NZ000093012
2	GISBORNE AERODROME	NZ000093292
3	PARAPARAUMU AWS	NZ000093417
4	INVERCARGILL AIRPOR	NZ000093844
5	RAOUL ISL/KERMADEC	NZ000093994
6	NEW PLYMOUTH AWS	NZ000933090
7	HOKITIKA AERODROME	NZ000936150
8	TARA HILLS	NZ000937470
9	CAMPBELL ISLAND AWS	NZ000939450
10	CHATHAM ISLANDS AWS	NZ000939870
11	AUCKLAND AERO AWS	NZM00093110
12	WELLINGTON AERO AWS	NZM00093439
13	KAIKOURA	NZM00093678
14	CHRISTCHURCH INTL	NZM00093781
15	ENDERBY ISLAND AWS	NZM00093929

Table 4.1.3: NZ Stations

4.1.4 Data Filtering and Preprocessing

Filtering:

- Select only New Zealand stations using "COUNTRY_NAME" == 'New Zealand' in enriched_stations dataset.
- Only keep ID, LATITUDE, LONGITUDE, and NAME columns.
- Joined selected stations to daily dataset using ID.
- Only TMIN and TMAX records filtered.

Storing:

- Save result as a Parquet file using .write().mode("overwrite") for faster processing.

Transformations:

- Add a YEAR column from the DATE.
- Pivot the data so that each row has both TMIN and TMAX.
- Final pivoted table consist of ID, DATE, NAME , YEAR, TMIN , TMAX
- Sort the data by station ID and date.

4.1.5 Individual Station Visualizations

To getting yearly temperature plots for each station:

Step	Description
Spark Data frame Convert to Pandas	Used .toPandas() for plotting graphs using Matplotlib.
Temperature conversion	TMIN and TMAX values is divide by 10 for convert from tenths of °C to actual °C.
Date formatting	Convert DATE to datetime using pd.to_datetime() and take year using .dt.year.
Group and average	Grouped by station and year, calculate average TMIN and TMAX per year.
Fill missing years	Create a full year range, fill missing years using NaN for consistent timelines.
Plot setup	Use plt.subplots() to create one plot per station, arranged in a grid layout.
Line colors	Use blue for average TMIN and red for average TMAX.
Axes standardization	Got consistent x-axis (year) and y-axis (temperature) using set_xlim(xmin, xmax) and set_ylim(ymin, ymax), based on the global min/max of year and temp.

Table 4.1.5: Steps to plot

This method helped me for clear comparison of long-term temperature patterns among all 15 stations in NZ.

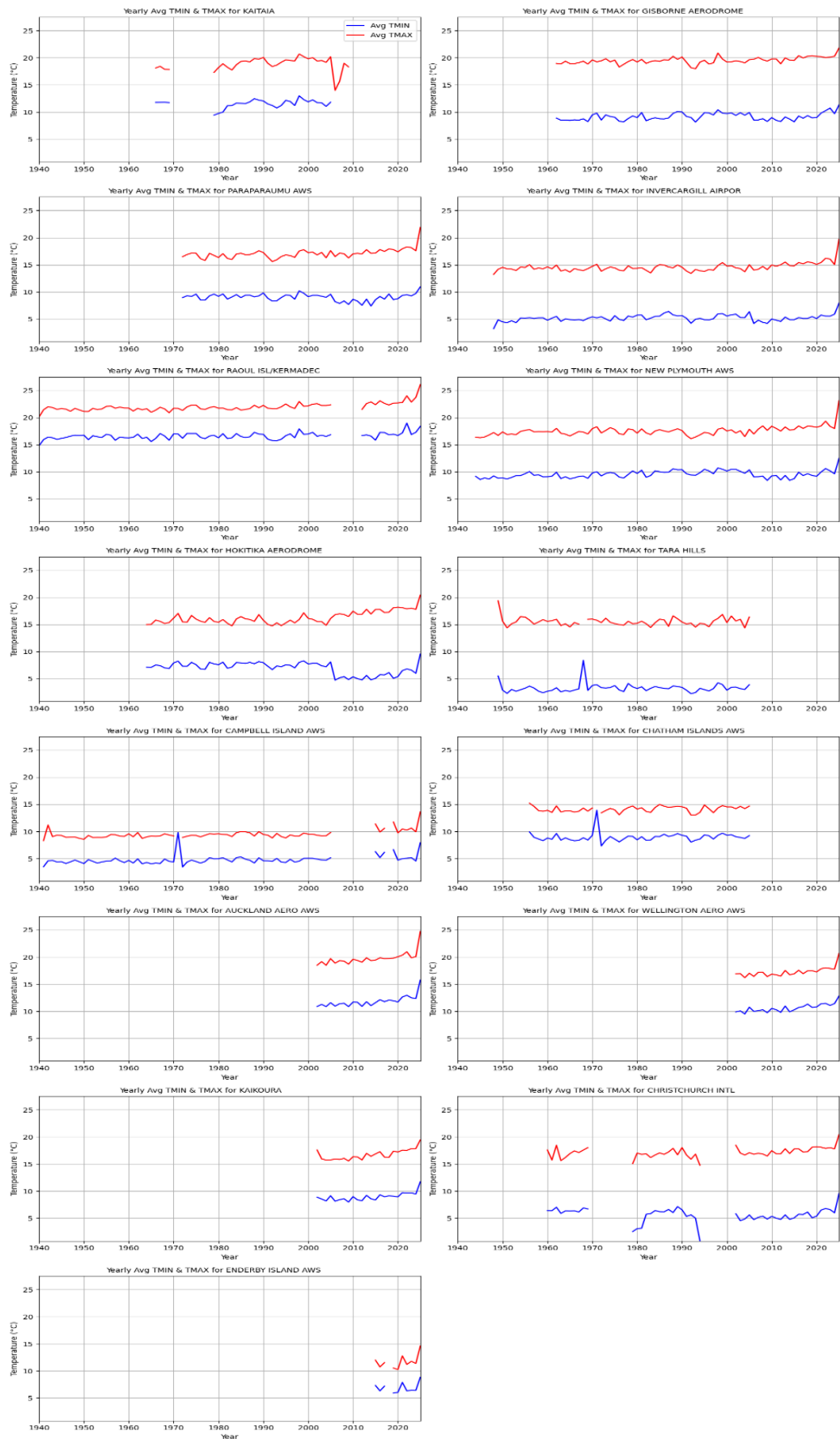


Figure 4.1.5: Yearly Temp Trends in NZ

Data Gaps and Station Coverage

Many stations have missing temperature data in some years, either due to late setup, early shutdown, or intermittent gaps. The table below summarizes missing years by station:

Station Name	No. of Missing Years
ENDERBY ISLAND AWS	76
AUCKLAND AERO AWS	62
WELLINGTON AERO AWS	62
KAIKOURA	62
KAITAIA	55
CHATHAM ISLANDS AWS	37
CHRISTCHURCH INTL	36
PARAPARAUMU AWS	32
TARA HILLS	30
HOKITIKA AERODROME	24
GISBORNE AERODROME	22
CAMPBELL ISLAND AWS	12
INVERCARGILL AIRPOR	8
RAOUL ISL/KERMADEC	6
NEW PLYMOUTH AWS	4

Table 4.1.5.1: Missing Years

Stations Likely Stopped Submitting Data

Station	Last Year of Data	Comment
KAITAIA	~2008	Data ends around 2008
TARA HILLS	~2005	No data after 2005
CHATHAM ISLANDS AWS	~2005	Data cuts off around 2005

Table 4.1.5.2: Stopping Stations

Recently Added Stations

Station	First Year of Data	Comment
ENDERBY ISLAND AWS	~2014	New station with limited data
KAIKOURA	~2002	Started recently with clean trend
AUCKLAND AERO AWS	~2002	Data begins around 2002
WELLINGTON AERO AWS	~2002	Data starts in early 2000s

Table 4.1.5.3: Added Stations

Notable Gaps in Long-Term Stations

Station	Missing Years	Comment
INVERCARGILL AIRPOR	1940–1947	Likely started recording in 1948
RAOUL ISL/KERMADEC	2006–2011	Temporary shutdown or data submission issue
NEW PLYMOUTH AWS	1940–1943	Station began reporting after 1943

Table 4.1.5.4: Data Gaps Stations

Data coverage varies across stations. These differences affect data quality and must be considered when interpreting long-term trends.

4.1.6 Temperature Trend Analysis by Station

For assessing long-term change in temperature, I calculated the yearly temperature trends in both minimum (TMIN) and maximum (TMAX) for each station:

Methodology

- For estimating temperature trends Linear regression was applied using `scipy.stats.linregress(YEAR, TMIN_C)` and `linregress(YEAR, TMAX_C)` (Cooper, n.d.).
- Rate of change in temperature over time (°C/year) is represented by slope.

Visualization

The below bar chart compares all 15 NZ stations TMIN and TMAX trends (in °C/year).

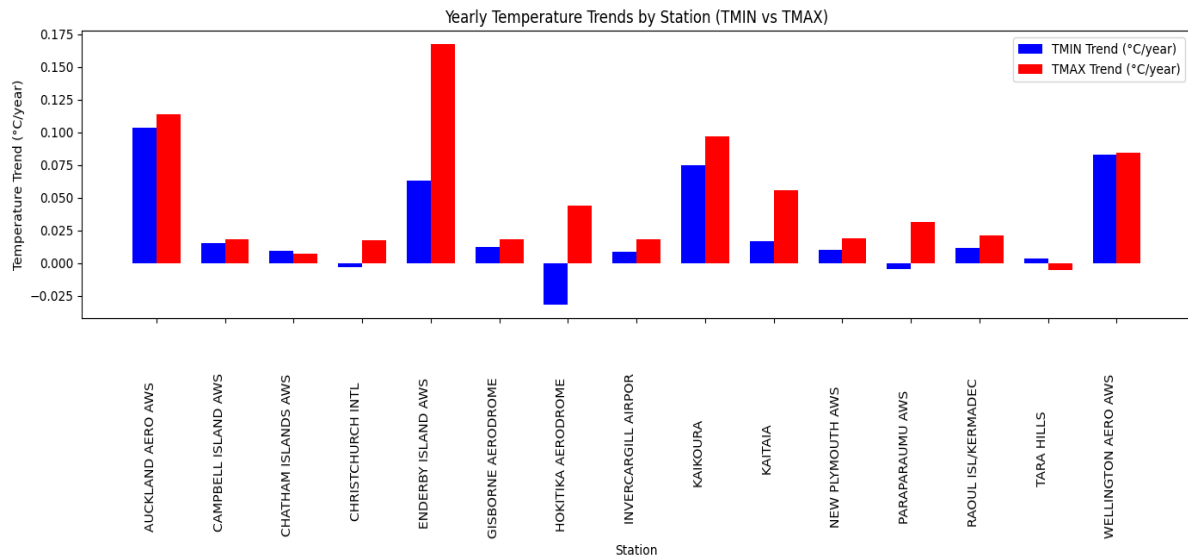


Figure 4.1.6: Yearly Temperature Trends by Station (TMIN vs TMAX)

Trend Type	Slope	Meaning
Positive	> 0	Temperature is rising
Negative	< 0	Temperature is falling

Table 4.1.6: Trend Meaning

Key Insights

- 11 out of 15 stations increase in both TMIN and TMAX, indicates warming pattern in New Zealand.
- Most stations slightly higher TMAX trends than TMIN, daytime temperature rise faster.
- A few stations (e.g., TARA HILLS (due to data), HOKITIKA, CHRISTCHURCH) show opposite trends, with one TMIN or TMAX is decreasing slightly. This may be due to local climate or data issues.
- No stations have both TMIN and TMAX decreasing, no stations overall cooling trends.
- Remote southern stations like Campbell Island, Chatham Islands, and Enderby Island also shows warming trend.

4.1.7 National Average Visualization

To show long-term temperature trends across New Zealand, I used the dataset was previously filtered and converted from Spark to Pandas after computing daily TMIN and TMAX per station.

4.1.7.1 Yearly Average TMIN & TMAX across New Zealand

Grouped the data by YEAR and calculated the national average TMIN and TMAX using `.groupby("YEAR").mean()`.

Matplotlib to create this line plot:

- Blue for average TMIN
- Red for average TMAX
- Common x-axis (year) and y-axis (°C)
- Gridlines and legend for clarity

This line plot shows the yearly average minimum (TMIN) and maximum (TMAX) temperatures for all New Zealand stations from 1940 to 2025. Early years like (1940-1950) show unusually high values due to data quality.

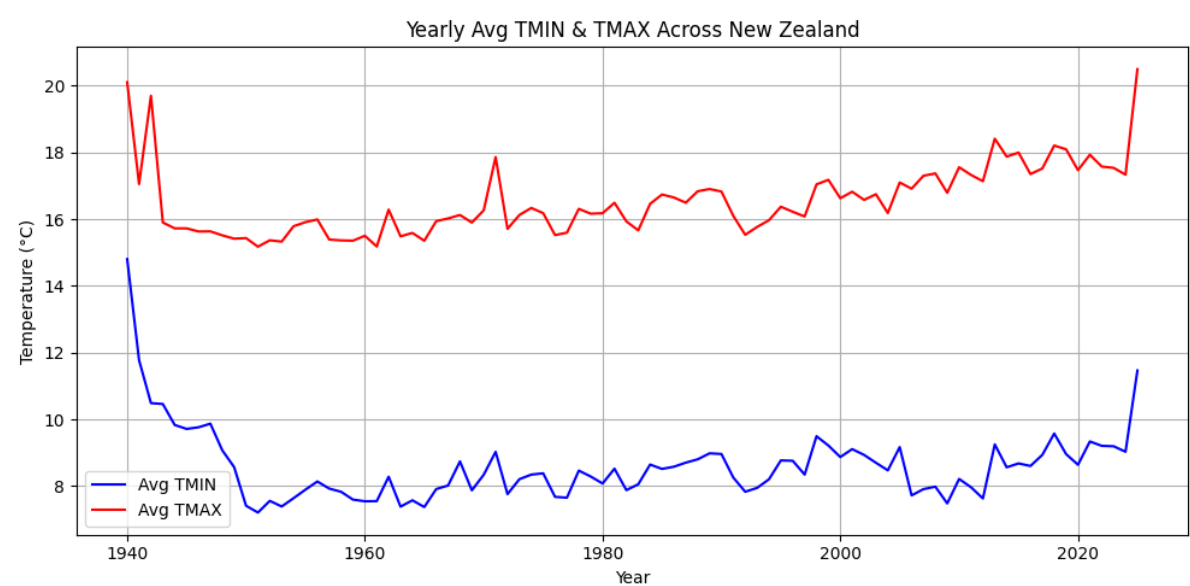


Figure 4.1.7.1: Yearly Average TMIN and TMAX Across New Zealand

Statistic	TMIN (°C)	TMAX (°C)
Minimum Value	7.21 (1951)	15.17
Maximum Value	14.80	20.49 (2025)
Long-term Average (1940–2025)	8.60	16.54
Average (Last 5 Years: 2021–2025)	9.65	18.17

Table 4.1.7.1: Summary of Yearly Average TMIN and TMAX (New Zealand)

This table showing increasing trends in both TMIN and TMAX, with the most recent years is the warmest one.

4.1.7.2 Seasonal Patterns of Temperature in New Zealand

To understand seasonal changes, I grouped the daily temperature data by month and calculated the average TMIN and TMAX for all stations in New Zealand.

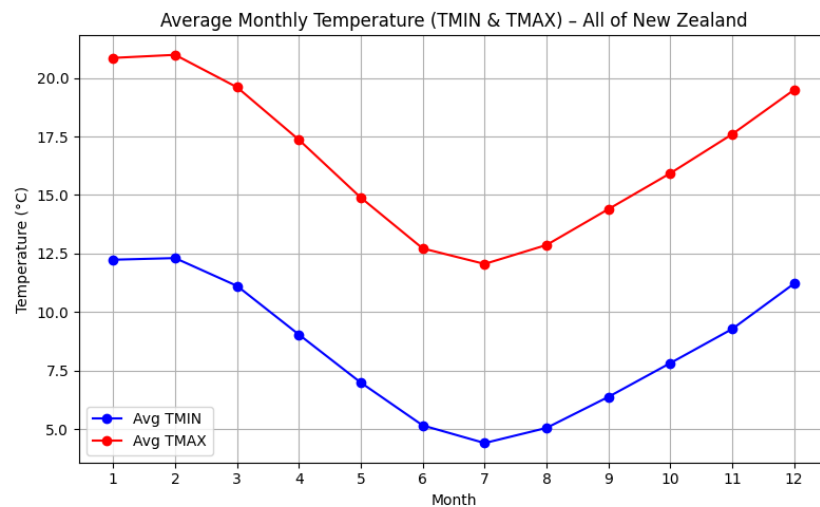


Figure 4.1.7.2: Seasonal Patterns of Temperature in New Zealand

The plot shows that:

- February is the warmest month, TMAX reaching 20.99 °C and TMIN around 12.31 °C.
- July is the coldest month, TMAX at 12.05 °C and TMIN at 4.40 °C.

These patterns reflect New Zealand's clear seasonal cycle, with warmer temperatures in summer and cooler ones in winter.

Statistic	TMIN (°C)	TMAX (°C)
Minimum Monthly Avg	4.40 (July)	12.05 (July)
Maximum Monthly Avg	12.31 (Jan/Feb)	20.99 (Feb)
Range	7.91	8.94
Overall Average (All Months)	8.41	16.56

Table 4.1.7.2: Summary of Monthly Average TMIN and TMAX - New Zealand

- On average, TMIN varies by 7.91 °C and TMAX by 8.94 °C throughout the year. Overall climate is mild and moderate
- Range shows how much temperatures change through the year. About 8 to 9° C, so seasons are noticeable but not extreme.

4.2 Global Precipitation Patterns

This section shows average yearly rainfall for each country using cleaned precipitation data. It includes a choropleth map highlighting global rainfall patterns in 2024.

4.2.1 Preprocessing and Aggregating Global Rainfall Data

- I filtered the daily dataset to include only PRCP (precipitation) records and added a YEAR column from the DATE.
- This was joined to station metadata for getting COUNTRY_NAME and COUNTRY_CODE.
- For cleaning the data, I removed all rows with non-null QUALITY_FLAG values.
- The QUALITY_FLAG column marks problems in the data. Only rows where QUALITY_FLAG was blank (null) is passed all quality checks. It makes results more accurate for analysis and plotting.
- Then data grouped by YEAR, COUNTRY_NAME, and COUNTRY_CODE and calculate average daily rainfall using `.groupBy().agg(avg("VALUE"))` also values converted from “tenths of mm” to “mm” by dividing by 10.
- The cleaned and aggregated average rainfall data saved as a Parquet file.
- For visualization, I loaded this file using Spark and then converted the data (avg_prdp) to a Pandas DataFrame (avg_prdp_pd) for plotting in Python.

4.2.2 Descriptive Statistics and Outlier Analysis

I first looked at basic statistics from the raw precipitation data. This showed missing values (like -9990) and very high numbers that may be errors. These checks helped us decide what to clean and remove.

Statistic	VALUE
Count	1,079,767,077
Mean	23.80
Standard Deviation	83.51
Minimum	-9990.0
Maximum	71374.0

Table 4.2.3.1: Initial Descriptive Statistics for PRCP Values

The dataset contains over 1.07 billion precipitation records, with an average daily rainfall of 2.38 mm. A high standard deviation (83.51) indicates large variation, might be due to extreme

events or errors. The minimum value (-9990) is a placeholder for missing data, while the maximum (7137.4 mm) may indicate an outlier.

Each observation in the dataset has a QUALITY_FLAG. A blank flag means the value passed. Non-blank flags indicate problems. For example:

- X: Value failed a bounds check (e.g., too high or too low)
- G: Gap in expected data
- O: Outlier based on climate patterns
- Other flags (e.g., S, I, D) also mark suspect data

Example flagged records:

ID	DATE	VALUE	COUNTRY	QUALITY_FLAG
USC00456534	1997-01-29	25146	United States	X
SA000040430	1997-01-07	7780	Saudi Arabia	G
PO000008522	1997-01-16	3520	Portugal	O

Table 4.2.3.2: Example of Flagged Records

Quality Flag Summary

The table below shows the count of precipitation records by QUALITY_FLAG value. Only records with NULL flags (i.e., passed all checks) were kept for analysis.

QUALITY_FLAG	Meaning (if known)	Count
NULL	Passed all checks	1,079,138,053
D	Duplicate check failed	362,164
L	Length check failed	160,414
K	Streak/frequent value failed	32,532
Z	Official Datzilla flagged	31,542
I	Internal consistency failed	15,428
S	Spatial consistency failed	14,959
O	Climatological outlier	8,964
G	Gap check failed	1,649
X	Bounds check failed	1,162
N	Naught check failed	210

Table 4.2.3.3: Count of Flagged Records

All records with non-null QUALITY_FLAG values were removed this include removed all negative values. This step eliminated 629,024 rows that failed checks for issues such as duplication, unrealistic values, or outliers.

Description	Count
Original Rows Count	1,079,767,077
Remaining Valid Data	1,079,138,053
Rows Removed During Cleaning	629,024

Table 4.2.3.4: Count of Valid Records

4.2.3 Descriptive Statistics of Aggregated Rainfall

I used the cleaned precipitation data to calculate average daily rainfall per country per year. This resulted in 17,710 country-year records. The descriptive statistics are shown below:

Statistic	AVG_PRCP (mm/day)
Count	17,710
Mean	3.96
Std Dev	6.00
Min	0.0
Max	224.47

Table 4.2.3: Descriptive statistics of PRCP Cleaned

The maximum value (224.47 mm/day) is likely an outlier. Many zero values were due to sparse data or actual lack of rainfall.

4.2.4 Outlier Investigation and Data Completeness

Most average rainfall values were reasonable mean is 3.96, some country-year records showed extreme or zero values. For instance, the maximum average was 224.47 mm/day, which is unusually high and likely an outlier. I also found **176 country-year pairs** with average rainfall of **0.0 mm/day**.

Upon further inspection, these zero values were mainly due to sparse records many had only 1-2 observations, not true absence of rainfall. For example:

Year	Country	Country Code	Average Rainfall (mm/day)	Record Count
1840	United States	US	0.0	1
1846	Netherlands	NL	0.0	1
1857	Croatia	HR	0.0	1
1857	Senegal	SG	0.0	30
1859	Senegal	SG	0.0	120
1860	Senegal	SG	0.0	91
1874	Norway	NO	0.0	1
1890	Turkmenistan	TX	0.0	2
1891	Tunisia	TS	0.0	2
1894	Tunisia	TS	0.0	2

Table 4.2.4: Example of Average PRCP equal Zero

This suggests the low values are due to limited data, not actual dry conditions. When interpreting historical rainfall trends this must be considered.

4.2.4.1 Global Rainfall Trends Over Time

This bar chart shows the average daily rainfall (in mm/day) for each year, based on data from all countries combined.

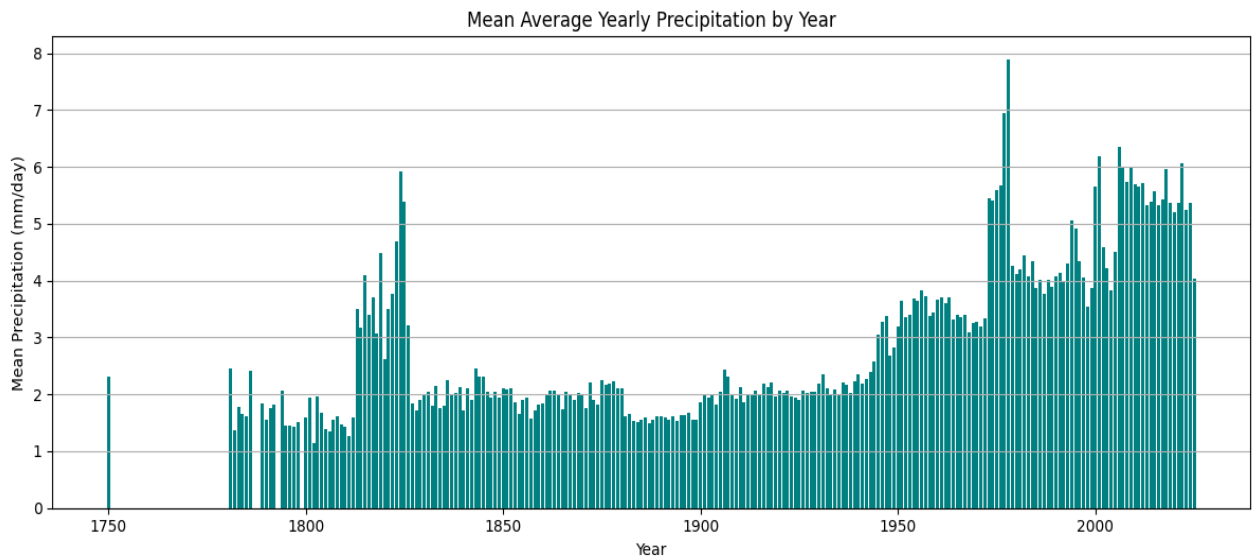


Figure 4.2.4.1: Mean average yearly precipitation across all countries (cleaned data).

- Before 1900 precipitation values are low and irregular, may be because of sparse data collection.
- From 1900 to 1950, gradual increases in rainfall averages, it reflects the rise of weather station networks.
- After 1950, values stabilize a little around 3-4 mm/day, indicates consistent global coverage.
- In 1958 and early 2000s sudden spikes might be because of extreme weather or bad data.
- This trend shows that when looking at global rainfall over time, we need to think about how much and where the data comes from.

4.2.5 Highest Average Rainfall in a Single Year

Belize in 1978 recorded the highest average daily rainfall at 224.47 mm/day.

Year	Country	Country Code	Avg PRCP (mm/day)
1978	Belize	BH	224.47

Table 4.2.5.1: Highest Rainfall Country

To verify if this is sensible, I checked the number of rainfall records for Belize in 1978.

DATE	VALUE (mm/day)
1978-06-16	490.0
1978-07-24	436.1

1978-09-18	396.0
1978-01-21	196.1
1978-01-23	53.1
1978-06-17	0.0
1978-07-25	0.0

Table 4.2.5.2: AVG PRCP values of Belize in 1978

After review found out:

- Only 7 records were available for Belize in 1978.
- Four extremely high daily values was recorded like 490.0 mm/day on June 16 and 436.1 mm/day on July 24.
- Other dates, like June 17 and July 25, recorded 0.0 mm/day, means inconsistent data.

These findings suggest the result may not be reliable due to:

- Sparse data
- Influence of Outlier
- Measurement or reporting problems

So, Belize shows highest average, but this number might not be accurate and should be consider carefully.

4.2.6 Visualizing Average Rainfall in 2024 Across Countries

I used a choropleth map to explore the global rainfall patterns. The different shades in map shows difference on average precipitation across countries.

Data Completeness and Quality Checks

After comparing with the full list of countries (218), found that 39 countries is missing from the 2024 data. This will cause gaps in choropleth.

Descriptive statistics for the available 2024 data (179 countries) showed:

Statistic	AVG_PRCP (mm/day)
Count	179
Mean	5.37
Std Dev	3.52

Min	0.13 (Qatar)
Max	18.05 (Angola)

Table 4.2.6.1: Descriptive Statistics of AVG PRCP Data 2024

To investigate extremes:

Angola has the highest average (18.05 mm/day), but it was based on only 2 records (36.1 and 0), makes it as an outlier.

Count	Mean	Std Dev	Min	Max
2	18.05	25.53	0	36.1

Table 4.2.6.2: Count of Angola Records

ID	DATE	ELEMENT	VALUE	COUNTRY_NAME
AO000066390	2024-04-10	PRCP	36.1	Angola
AO000066390	2024-04-11	PRCP	0.0	Angola

Table 4.2.6.3: Only Two Records for Angola

Qatar got the lowest average (0.13 mm/day), this is based on 76 records, makes it's more reliable.

Count	Mean	Std Dev	Min	Max
76	0.129	0.529	0.0	3.3

Table 4.2.6.4: Count of Qatar Records

Distribution of Average Rainfall in 2024 by Country

The distribution of average daily rainfall across 179 countries in 2024 is shown in this histogram. Most of the countries are in between 2 mm/day and 7 mm/day, highest number of countries around 2.5-5 mm/day. A few countries got over 15 mm/day, which shows outliers or extreme events.

This right skewed distribution means, high rainfall is less common. Some countries rainfall close to 0 mm/day, which indicates very dry climates or limited data.

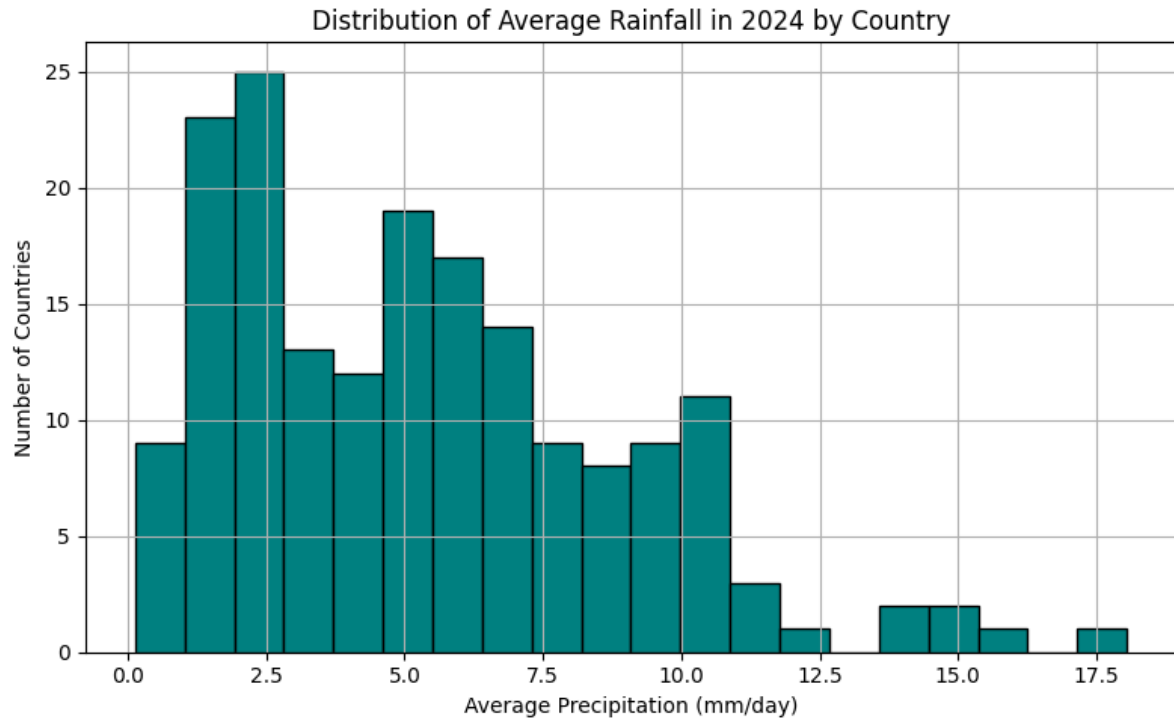


Figure 4.2.6.1: Histogram of average daily precipitation (mm/day) in 2024 per country

Following the histogram, the boxplot below gives another view of how average rainfall was spread across countries in 2024.

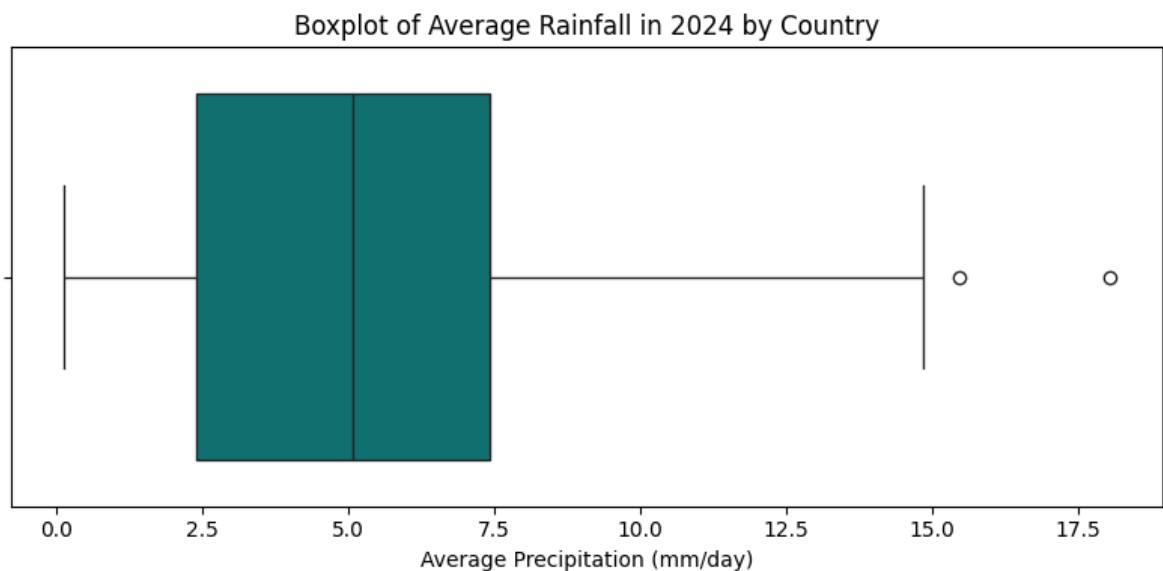


Figure 4.2.6.2: Boxplot of Average Daily Rainfall (2024) by Country

In this graph clearly shows most countries had rainfall between 2.5 and 7 mm/day, and the middle value (median) was around 5 mm/day. Two dots on the right are outliers.

To identify countries with unusually high or low rainfall, I used the Interquartile Range (IQR) method.

- Q1 (25th percentile) and Q3 (75th percentile) used to calculate the IQR.
- Countries with values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were flagged as outliers.

Using the IQR method, two countries are outliers:

- Angola (18.05 mm/day) - already identified with only 2 records.
- Guinea (15.48 mm/day) - very high rainfall, 552 Observations, it is a valid one.

Top 10 Rainiest Countries in 2024

Below bar chart shows countries had highest average daily rainfall in 2024. Angola ranked first, followed by Guinea, Gambia, and Indonesia. Most of these (Indonesia, Guinea, The Gambia, Vanuatu, Brunei, Costa Rica, India, and American Samoa) countries are in tropical or equatorial regions, naturally receive heavy rainfall (Wikipedia, 2025).

However, some results (like Angola) may be affected by limited data.

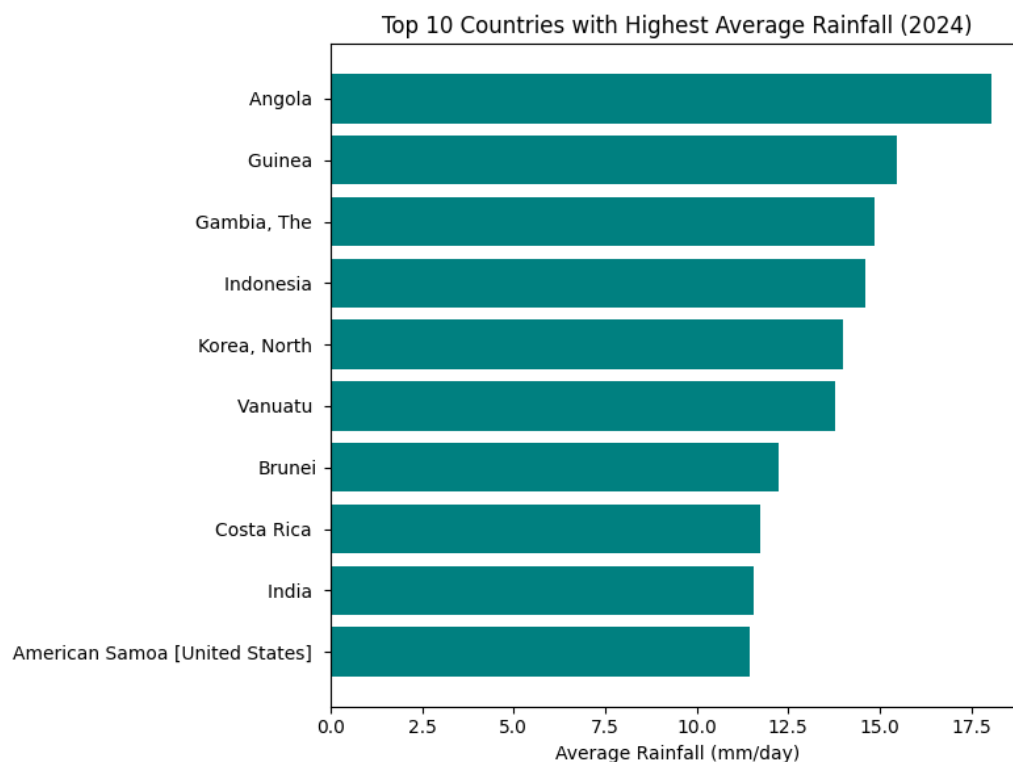


Figure 4.2.6.3: Top 10 countries by average precipitation (mm/day) in 2024.

COUNTRY_NAME	NUM_OBS
India	15,863
American Samoa [United States]	1,097

Vanuatu	1,057
Indonesia	783
Guinea	552
Costa Rica	459
Korea, North	315
Brunei	267
Gambia	64
Angola	2

Table 4.2.6.5 : Observation Counts for Top 10 Countries

Bottom 10 Driest Countries (2024)

The below bar chart shows countries got the lowest average daily rainfall in 2024. Qatar had the lowest value, followed by Iraq, Tunisia, and Chile. Most of these countries (Qatar, Iraq, Tunisia, Algeria, Morocco, Iran, Cyprus, and Chile) are in arid or desert regions have relatively low rainfall. These pattern consistent with global patterns of precipitation.

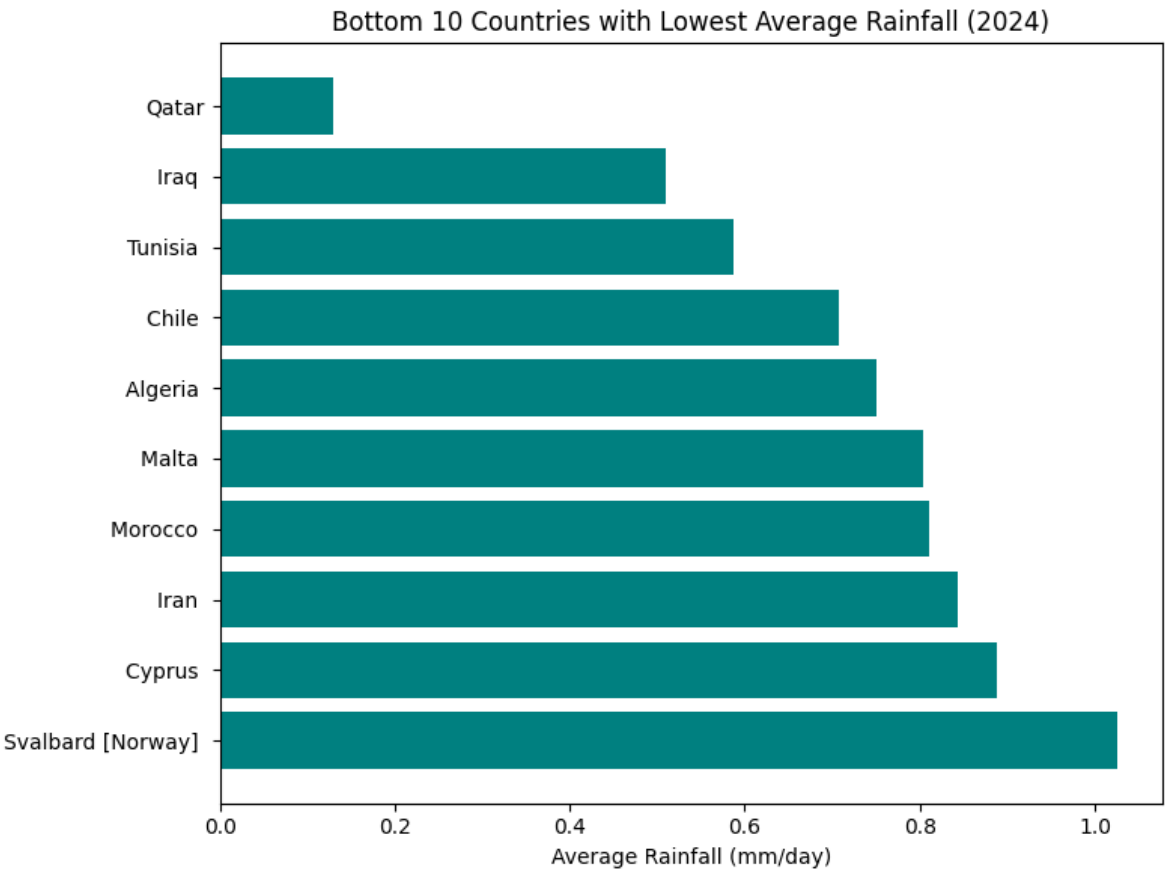


Figure 4.2.6.4: Bottom 10 countries by average precipitation (mm/day) in 2024.

COUNTRY_NAME	NUM_OBS
Algeria	13,975
Iran	10,247
Chile	5,243
Morocco	3,713
Tunisia	1,669
Svalbard [Norway]	666
Malta	350
Cyprus	348
Iraq	278
Qatar	76

Table 4.2.6.6: Observation Counts for Bottom 10 Countries

Even if observation counts variation exists, countries like Indonesia, Guinea, The Gambia, Vanuatu, Brunei, Costa Rica, India, and American Samoa are the rainiest tropical/equatorial countries, which follows global climatic pattern. The arid and desert countries like Qatar, Iraq, Tunisia, Algeria, Morocco, Iran, Cyprus, and Chile are driest countries follows the same. These make some confidence in the overall dataset structure.

4.2.7 Visualizing 2024 Rainfall - Choropleth Map

I used Plotly to generate choropleth map. Country names were cleaned by regular expressions for removing bracketed text , (for e.g. [United States], (Territory)) and ISO-3 codes was assigned using **coco** library (Cohen, 2022).

The map built using `px.choropleth()` with ISO-3 codes for locations, rainfall values (AVG_PRCP) for coloring, and “Turbo” as the color scale.

After assigning ISO-3 codes,

The following country names did not match automatically and need manually assigned ISO-3 code:

Jan Mayen, Midway Islands, Virgin Islands, and Wake Island.

I checked that all countries had valid codes after this step.

Out of 179 countries in the dataset, **177 unique ISO-3 codes** used for the map. Two duplicate codes are found:

- SJM: Jan Mayen and Svalbard
- UMI: Midway Islands and Wake Island

These small regions shared the same ISO3 code.

Average Rainfall by Country in 2024

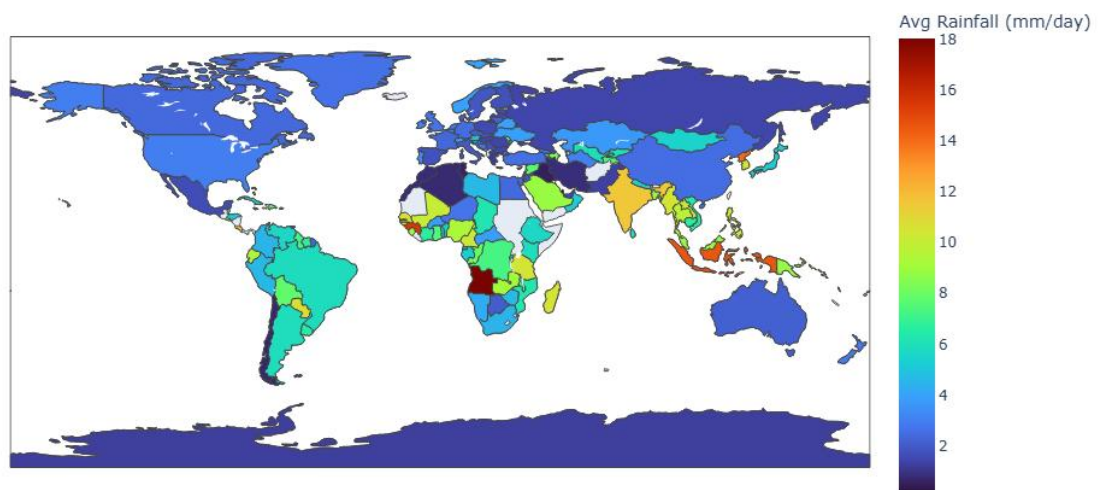


Figure4.2.5: Average Rainfall by Country in 2024

This choropleth map shows the average daily precipitation (mm/day) across 177 countries in 2024. Darker red/orange colours indicate higher rainfall. Regions like Central Africa and Southeast Asia show the highest values, while drier areas like North Africa (e.g., Algeria, Libya, Egypt), the Middle East (e.g., Qatar, Iraq, Iran), and Central Asia appear in blue shades. Most countries experienced moderate rainfall with colour shades green to light blue (Mongolia., New Guinea).

5. Conclusions

In this assignment, a large-scale climate data I successfully processed, analysed, and visualized using Apache Spark and Python-based visualization tools. The project needs working on both structured and fixed width formatted data and, I faced challenges of working with big datasets, like memory management, file parsing, and usage of efficient computation spark operations.

Main Outcomes from this assignment are:

- An enriched station metadata table made by joining station, state, country, and inventory data.
- Data coverage issues identification, such as stations reported TMAX but not TMIN, and detected updating gaps in the metadata.
- Analysis made on station counts, network membership, hemisphere distribution, and territory coverage.
- Geographic distances computed using user-defined functions and found closest stations in New Zealand.
- By aggregating data and then plot temperature trends in New Zealand stations, warming trends in both TMIN and TMAX are revealed.
- By global precipitation analyses, identify countries with the highest and lowest average rainfall and created maps and bar charts to visualize the results.
- Data issues like missing values, outliers, and inconsistent country naming are found.

Overall, this assignment helps to understand of big data processing using Spark, also improved skills in data cleaning, transformation, exploratory analysis, and geospatial/time-series visualization. It emphasizes value of critical thinking and verification when interpreting environmental data at a global scale.

6. References

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7. Appendices

Figure B1: Monthly Average Temperatures for New Zealand Stations

This figure shows average monthly TMIN (blue) and TMAX (red) for each station. It highlights seasonal patterns, with warmer temperatures in summer and cooler in winter. More details are in Table A1.

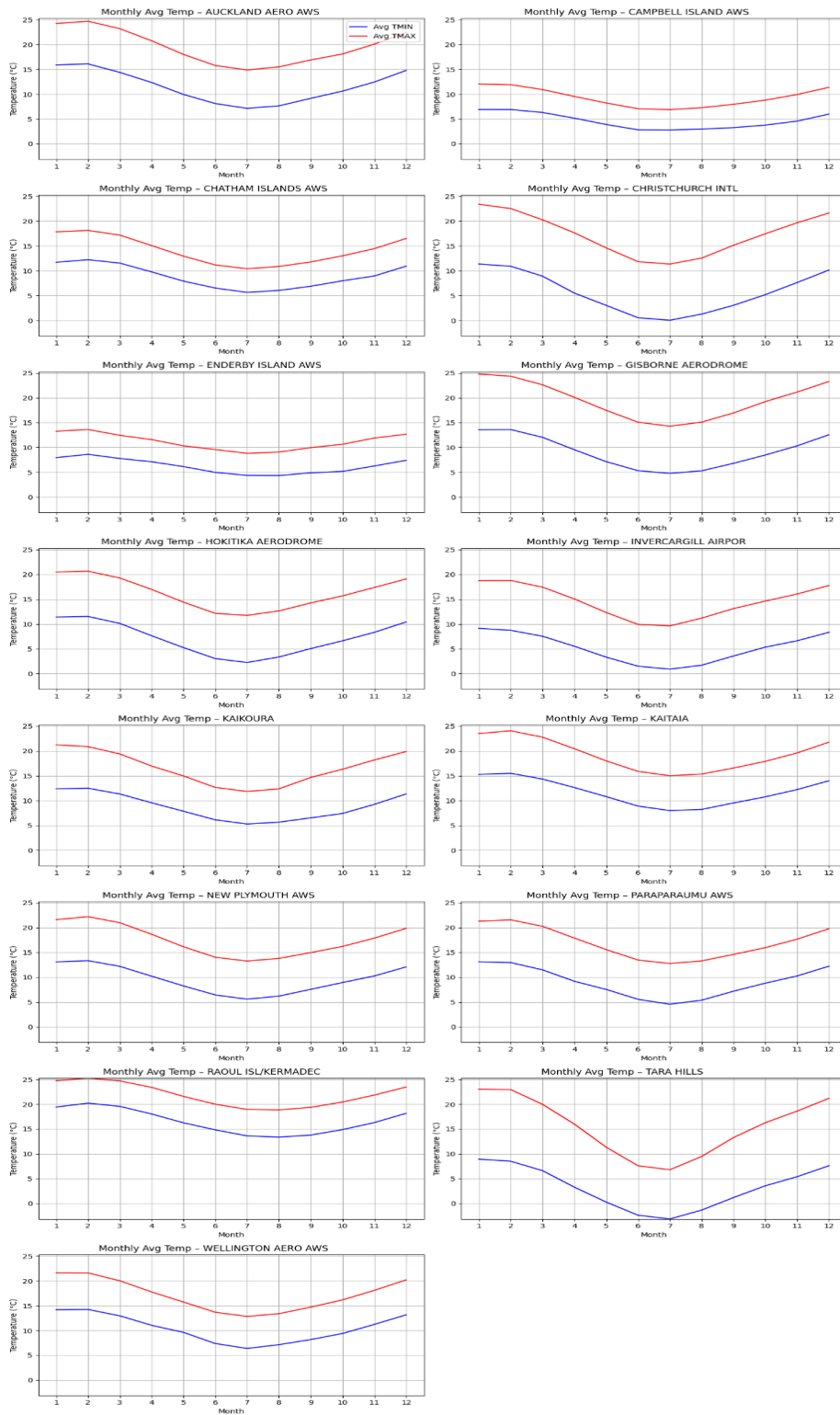


Table A1: Monthly Temperature Summary by Station

Shows the warmest and coldest months, temperature ranges, and seasonal insights for each station.

Station Name	Warmest Month	Coldest Month	TMAX Range (°C)	TMIN Range (°C)	Annual Temp Range (°C)	Insight Summary
KAITAIA	Feb	July	15 - 24	8 - 16	16.1	Warm summers, mild winters, moderate variation
GISBORNE AERODROME	Jan	July	14 - 25	5 - 14	20.0	Hot summers, cool winters, high variation
PARAPARAUMU AWS	Feb	July	13 - 22	5 - 13	17.0	Warm summers, mild winters, moderate variation
INVERCARGILL AIRPOR	Feb	July	10 - 19	1 - 9	17.9	Cool summers, cold winters, moderate variation
RAOUL ISL/KERMADEC	Feb	Aug	19 - 25	13 - 20	11.9	Warm all year, mild seasonal change
NEW PLYMOUTH AWS	Feb	July	13 - 22	6 - 13	16.6	Warm summers, mild winters, moderate variation
HOKITIKA AERODROME	Feb	July	12 - 21	2 - 12	18.4	Mild summers, cool winters, moderate variation

TARA HILLS	Jan	July	7 - 23	-3 - 9	26.2	Hot summers, very cold winters, very high variation
CAMPBELL ISLAND AWS	Jan	July	7 - 12	3 - 7	9.3	Mild all year, very low variation
CHATHAM ISLANDS AWS	Feb	July	10 - 18	6 - 12	12.5	Mild summers and winters, low variation
AUCKLAND AERO AWS	Feb	July	15 - 25	7 - 16	17.6	Warm summers, mild winters, moderate variation
WELLINGTON AERO AWS	Jan	July	13 - 22	6 - 14	15.2	Warm summers, mild winters, moderate variation
KAIKOURA	Jan	July	12 - 21	5 - 13	16.0	Warm summers, cool winters, moderate variation
CHRISTCHURCH INTL	Jan	July	11 - 23	0 - 11	23.4	Hot summers, cold winters, high variation
ENDERBY ISLAND AWS	Feb	Aug	9 - 14	4 - 9	9.3	Cool all year, very low variation

Figure B2: Temperature Distribution Histograms by Station

This figure displays temperature distributions for TMIN and TMAX at each New Zealand station. The histograms shows how frequently different temperature values occur and highlight station-specific climate characteristics and data quality.

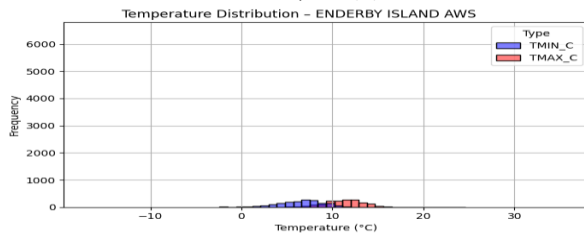
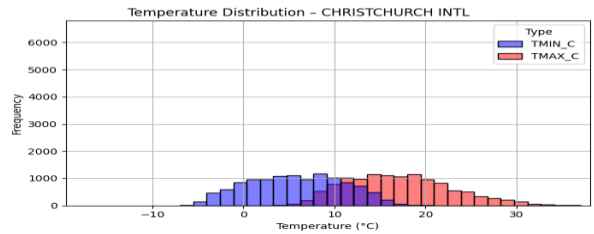
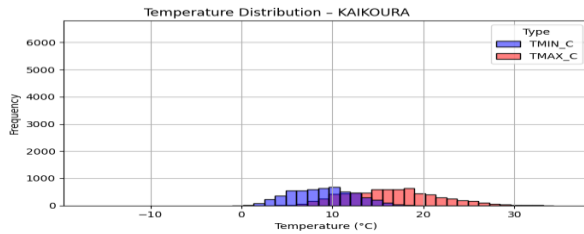
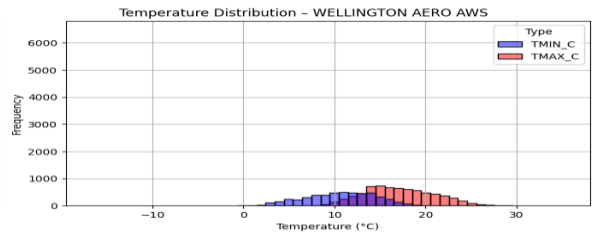
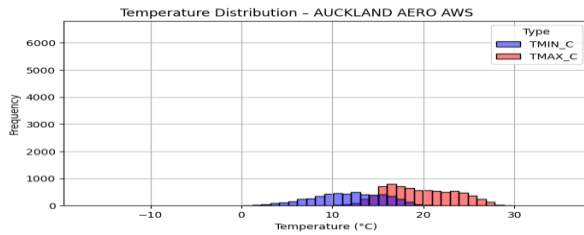
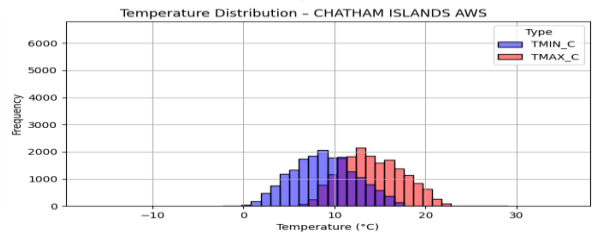
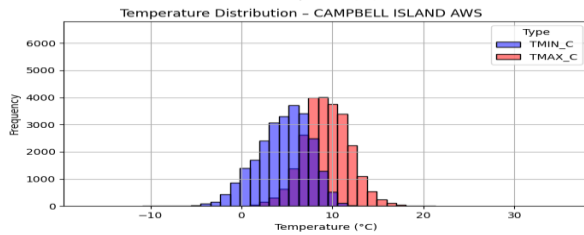
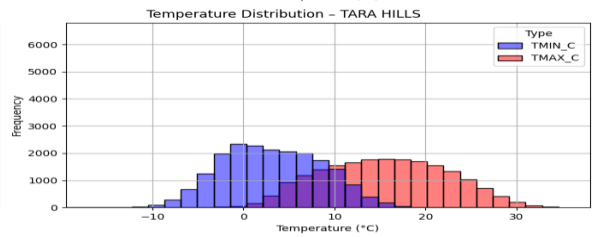
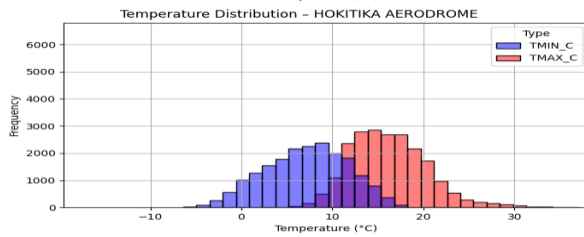
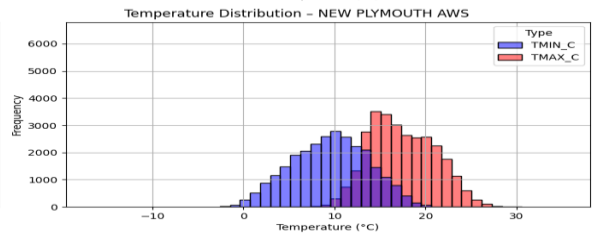
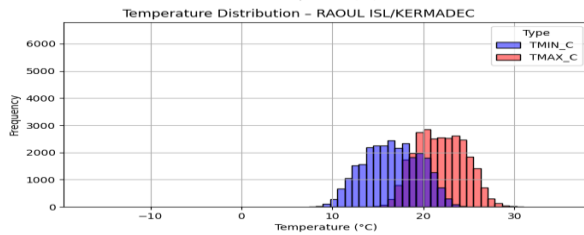
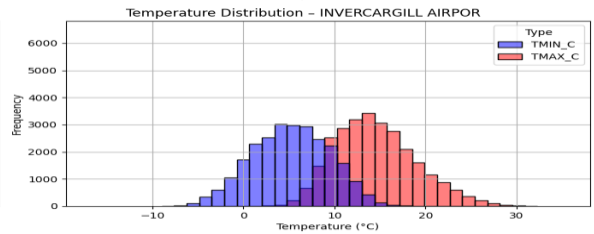
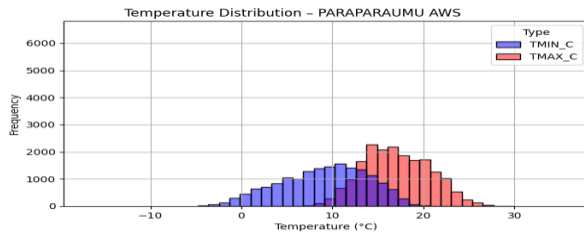
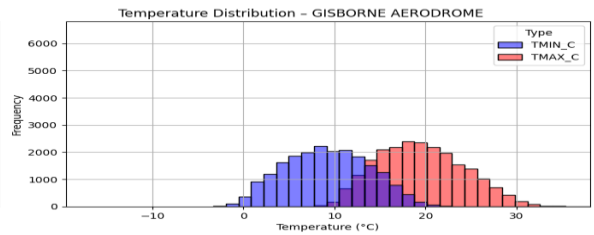
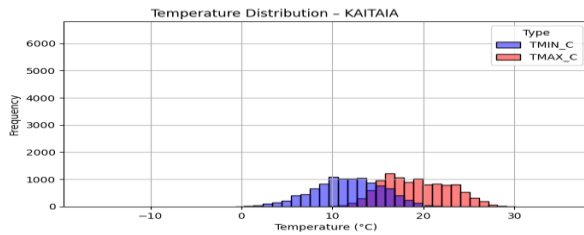


Table A2: Temperature Distribution Summary by Station

Station Name	Observation Count	Distribution Comment
RAOUL ISL/KERMADEC	28,030	Smooth, bell-shaped curves with good coverage
NEW PLYMOUTH AWS	29,469	Balanced distribution with widespread
INVERCARGILL AIRPOR	27,834	Well-shaped with moderate skew and wide TMAX spread
CAMPBELL ISLAND AWS	25,513	Narrow, centered curves; low variation
GISBORNE AERODROME	22,768	Centered peaks; moderate variation
HOKITIKA AERODROME	22,075	Broad spread; TMIN shows more cold extremes
TARA HILLS	19,760	Widespread; large range from cold to hot
PARAPARAUMU AWS	19,176	Symmetric shape with moderate spread
CHATHAM ISLANDS AWS	17,490	Narrow, centered shape; mild and stable climate
CHRISTCHURCH INTL	16,244	Wide tails in both TMIN and TMAX; strong seasonal variation
KAITAIA	11,046	Slightly skewed; warmer bias, limited recent data
AUCKLAND AERO AWS	8,059	Narrow spread; short duration but clear separation
WELLINGTON AERO AWS	7,855	Moderate spread; recent and consistent data
KAIKOURA	7,793	Narrow shape; limited data but stable pattern
ENDERBY ISLAND AWS	2,093	Sparse data; limited variation and distribution shape

Figure B3: National Temperature Distribution Summary - All NZ Stations

Histogram of TMIN (blue) and TMAX (red) for all observations in New Zealand. Shows how temperatures are distributed overall.

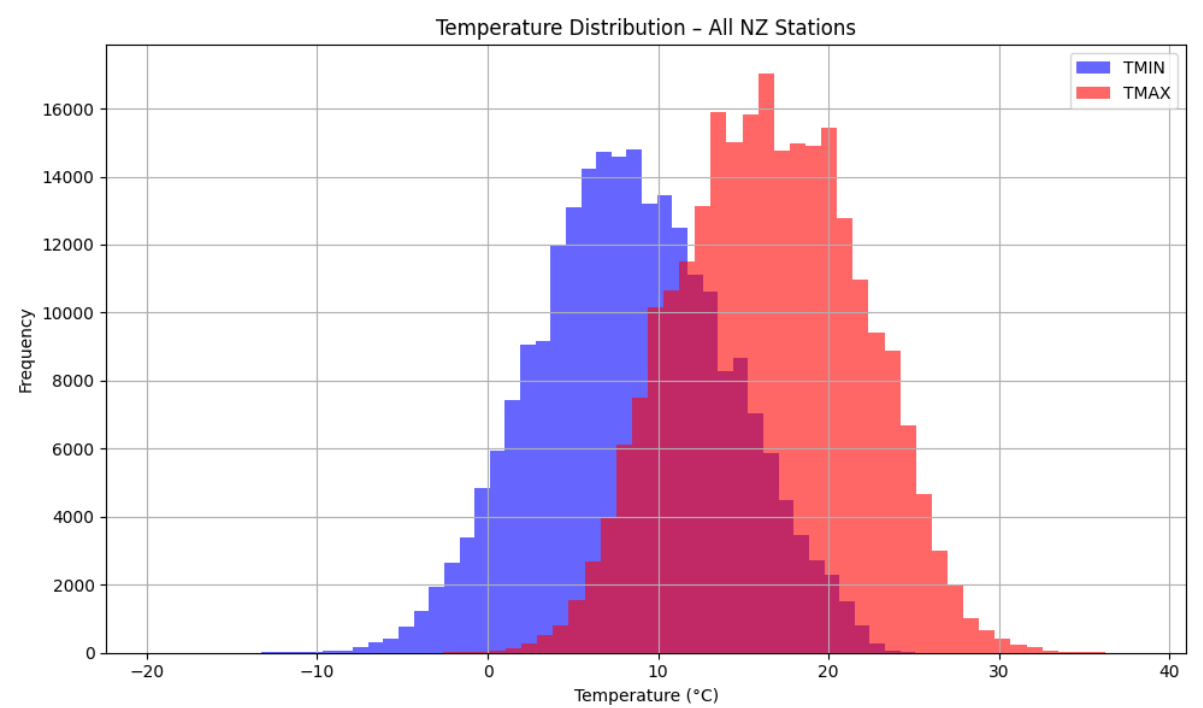


Table A3: Distribution Summary Statistics of TMIN and TMAX

Statistic	TMIN (°C)	TMAX (°C)	Comment
Count	237,320	254,121	Number of valid temperature records used in the analysis.
Mean	8.44	16.53	The average minimum and maximum temperatures across all stations.
Standard Deviation	5.57	5.30	Shows how much temperatures vary from the average moderate variation.
Minimum	-19.50	-8.20	Coldest recorded temperatures; may indicate rare extremes. Outliers
25th Percentile	4.50	12.70	25% of values fall below this temperature - represents cooler days.
Median	8.30	16.50	The middle value - 50% of days are warmer, 50% are cooler.
75th Percentile	12.40	20.40	75% of values fall below this - represents relatively warmer days.

Maximum	25.10	38.10	Hottest recorded temperatures across all stations.
Mode	7.80	15.00	Most frequently occurring temperature